

Active Maintenance Report: coppock_kaur_2022

Coppock and Kaur (2022, American Journal of Political Science): Qualitative
Imputation of Missing Potential Outcomes

2026-08-03

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This repository holds the actively maintained replication code for Coppock and Kaur (2022), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi [10.1017/S1049096521000366](https://doi.org/10.1017/S1049096521000366)) to a set of published archives.

Article	10.1111/ajps.12697
Replication archive	10.7910/DVN/2IVKXD

The data are not redistributed here. The deposit is 10 files and about 0.7 MB, and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the exhaustive list of numbers the article and its supporting information print, the ground truth CSV is the comparison against the deposit and against the rewrite, and `build_ground_truth.R` is the gate that stops the run if a published number is checked by neither instrument. `coppock_kaur_2022_errata.pdf` at the root lists the sentences in the article that its own tables contradict. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See LICENSE.

To reproduce. Clone or download the repository, open `coppock_kaur_2022.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 10 files, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, janitor, here, and for this report knitr and kableExtra. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about a minute, most of it in the twenty-seed sensitivity check at the end.

A successful run overwrites `maintained/output/`, which is committed: **git diff on that folder is the reproduction check**. The figure PDFs always differ, because a PDF records the time it was written; the CSV and PNG output comes back byte-identical.

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Yes. All five analysis scripts and the helpers file execute without error on R 4.6.0. Every package the archive names still installs from CRAN: `tidyverse`, `xtable`, `janitor` and `reshape2`. Nothing in the archive has been removed from the language; what has accumulated is deprecation. `geom_errorbarh()` and the `<ggplot> %+% data` idiom were deprecated in `ggplot2` 4.0.0, the `size` aesthetic for lines in 3.4.0, and `reshape2::melt` and `dcast` are superseded by `tidyr`. All of them still work and all of them warn. Everything that stands between this archive and a clean run is resolvable by substitution.

Running is not the same as reproducing, and one thing keeps the archive from reproducing its own figures exactly. `figures_2_3_A1_A2.R` estimates the bounds by simulation, 1,000 draws for Figures 2 and A1 and 10,000 for Figures 3 and A2, and it sets no seed. The published figures are labelled with rounded percentage points, so most of the simulation noise is absorbed by the rounding, but not all of it: repeating the Figure 2 and A1 simulation at twenty seeds moves at least one of the 28 rounded labels off its published value in 9 of them. A reader running the deposited script today has close to even odds of seeing a figure that disagrees with the article somewhere, with no error and no warning. The maintained rewrite fixes the seed at 12345, which reproduces all 28 published labels of Figures 2 and A1 and all 56 of Figures 3 and A2.

Two smaller things. `table_4.R` prints its bounds as fractions where the paper prints percentage points, so the deposited output reads `[0, 0.3]` against the article's `[0, 30]`. It is a display inconsistency and not a numerical one. And the writing of every table and figure is commented out: each script builds the object and prints it, but the `xtable` and `ggsave` calls that would put it on disk are behind `#`. The archive is a set of scripts that show their work rather than a pipeline that produces the article's floats.

1.2 Does the maintained rewrite reproduce the paper?

Yes. 217 of the 227 verifiable ground truth claims match the published values to reported precision: every cell and bound of Table 3, every bound of Figure 1, every row of Table 4, the democratization row of Table 5, all 28 labels of Figures 2 and A1, all 56 of Figures 3 and A2,

all ten bounds of Figure 4, 761 of the 763 cells the two appendix case tables print, and every in-text quantity from the abstract, the Expert Survey section and Appendix A.

The 10 that do not match fall into three groups. Seven are the five sentences in the article that the article’s own tables contradict, two of the five stating a pair of numbers rather than one; the rewrite agrees with the tables in every case, and all five are set out in the errata section below and in `coppock_kaur_2022_errata.pdf`. Two are cells of appendix Table B.2, one a probability the appendix contradicts elsewhere and one an observed outcome the deposited data record differently. The last is a probability the article states twice in prose and the deposited data carry as its complement. The remaining 5 recorded quantities have no comparison to make: four are the end-of-conflict row of the imputation summary, which the rewrite computes and neither the article nor the appendix prints, and one is a survey fact the deposit never recorded.

2 Paper overview

Citation: Coppock, A. and Kaur, D. (2022). “Qualitative imputation of missing potential outcomes.” *American Journal of Political Science*, 66(3), 681-695. DOI: 10.1111/ajps.12697

Summary: The paper proposes a framework for meta-analysing qualitative causal inferences. A causal claim about a single case is a claim about the value of a missing potential outcome, so a set of such claims can be treated as imputations and fed into Manski (1999) extreme value bounds, which are the logical range of average causal effects consistent with what has been observed. Bounds on a binary outcome start 200 percentage points wide and narrow to 100 once the world reveals half the potential outcomes; each qualitative imputation narrows them further, and cases too murky to call are left unimputed rather than guessed. Where the analyst holds a probability rather than a point belief, the extension enumerates the 2^k consistent worlds and takes the probability-weighted bounds. The application is to 63 cases eligible for a transitional truth commission upon democratization, eight of which received one, with authoritarian resumption within ten years as the outcome. Imputation in four steps takes the bounds on the ATE to $[-2, 49]$, 51 points wide. An email survey of country experts, 20 of whom responded, supplies an independent set of imputations; deferring to the experts where they answered narrows the bounds to 44 points. Appendix A repeats the exercise for 54 end-of-conflict cases.

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
--------	---------------------	------------

quimpo_helpers.R	Clean (sourced by the analysis scripts)	%>% to <code>&#124;></code> , <code>map_df()</code> to <code>map()</code> + <code>bind_rows()</code>
table_3_figure_1.R	Runs; deprecation warning	<code>size=</code> to <code>linewidth=</code>
table_4.R	Runs clean	Multiply the bounds by 100 to print percentage points, as the paper does
table_5.R	Runs clean	<code>recode_factor()</code> to <code>case_when()</code> + <code>factor()</code>
figures_2_3_A1_A2.R	Runs; three deprecation warnings; unseeded simulation	<code>melt/dcast</code> to <code>pivot_longer/pivot_wider</code> ; <code>geom_errorbarh()</code> to <code>geom_errorbar(orientation='y')</code> ; <code>%+%</code> to a named plot function; <code>do()</code> to <code>reframe()</code> ; <code>set.seed(12345)</code>
figure_4.R	Runs; deprecation warning and a <code>melt()</code> id-variable warning	<code>melt</code> to <code>pivot_longer</code> with an explicit <code>names_to</code> ; <code>size=</code> to <code>linewidth=</code>

3.1 The unseeded simulation

The probabilistic extension is what makes Figures 2, 3, A1 and A2 stochastic. Where the authors hold a probability rather than a point belief about a missing potential outcome, `sample_bounds()` draws binary potential outcomes from those probabilities and recomputes the bounds, and the reported estimate is the mean over draws. That is a design decision the paper explains and the rewrite keeps. What the archive omits is `set.seed()`, so the figures are one unlabelled draw.

Table 2: Repeating the Figure 2 and A1 simulation at twenty seeds. Of the 28 rounded labels those two figures carry, the number that move off the value the rewrite commits, which is also the value the published figures carry.

Rounded labels moved	Seeds
0	11
1	6
2	3

The size of the disturbance is one percentage point in the rounded label, never more, and it never touches the substantive claims: the final bounds are $[-2, 49]$ at every seed tested. The point is not that the archive gets a different answer but that it gets a slightly different figure each time it is run, and it offers a reader no way to tell whether a discrepancy is noise or a mistake.

3.2 Checksums

Table 3: The deposit as Dataverse serves it with `?format=original`. All ten files match the checksums Dataverse publishes, and all ten match the copy this program downloaded in March 2026.

File	Bytes	MD5 served (first 12)	Published MD5 agrees
cases_clean.csv	21433	0a238c073307	yes
codebook_cases_clean.pdf	348561	87154446febf	yes
figure_4.R	4282	d62dbb3dc25a	yes
figures_2_3_A1_A2.R	5859	b06b1c39bfba	yes
logfile.html	278521	831c13a4b09c	yes
quimpo_helpers.R	1517	876ec89e03d6	yes
README.txt	1712	634560b77b47	yes
table_3_figure_1.R	3456	e8fb68e7b478	yes
table_4.R	2220	237c27908efb	yes
table_5.R	1184	fd0efb3ad0e6	yes

Harvard Dataverse ingests `cases_clean.csv` as tabular data and derives a `.tab` representation from it, so the manifest records `originalFileName` and `originalFileSize` rather than the derived names and sizes, and `download_original.R` asks for `?format=original`. The published MD5 describes the deposited bytes, not the derived ones, and here it verifies them exactly. Other deposits in this program carry published checksums that verify neither file, so the download script checks against the served checksum and reports any published disagreement rather than failing on it.

4 Errata

Nothing in the deposited code is wrong. Five claims in the article’s prose are, each contradicted by a table in the same article, and the maintained rewrite therefore disagrees with the sentence and agrees with the float. All five are set out with the corrected sentence in `coppock_kaur_2022_errata.pdf` at the root of this repository. None of them changes a conclusion.

The toy example’s treated outcomes. The text introducing the toy example says “the outcome for three of the treated units and one of the untreated units is 1”. Table 3 on the facing page shows units 1 through 4 treated with an observed outcome of 1, which is four, and the deposited `Y <- rep(c(1, 0, 1, 0), c(4, 3, 1, 2))` encodes four. Four is also what the bounds require: the initial bounds of $[-40, 60]$ follow from four treated ones and would be $[-50, 50]$ with three.

The probabilistic extension’s probabilities. The text says that instead of imputing 0 for units 6 and 7, “we think the probabilities of being a ‘1’ for units 6 and 7 are .3 and .4”. Table 4 gives the four scenario probabilities as 0.56, 0.14, 0.24 and 0.06, which is the product distribution of 0.2 and 0.3, not of 0.3 and 0.4, which would give 0.42, 0.18, 0.28 and 0.12. The deposited `table_4.R` uses 0.2 and 0.3 and reproduces Table 4 exactly, as does the rewrite.

Appendix A’s bounds width. Appendix A says the imputations shrink “the width of the extreme value bounds from 100 to 41 points” and then, in the next sentence, that “the bounds come to $[-3, 42]$ (only 45 points wide)”. Both cannot hold. The pipeline gives final

end-of-conflict bounds of $[-2.9, 41.5]$, a width of 44.4 points, which rounds to 44 and which is 45 if taken as the difference of the rounded endpoints, as the second sentence does. The 41 matches nothing.

How the 63 missing potential outcomes were imputed. The empirical section says “for all 63 unobserved potential outcomes, we imputed 5 with certainty, 26 probabilistically and we left 32 unimputed”. An imputation is made with certainty when the stated probability that the missing potential outcome equals 1 is exactly 0 or exactly 1. Appendix Table B.2 shows four such cases, Brazil, Uruguay and two Thailand transitions, and 27 whose probability lies strictly between; the deposited data agree. The 32 left unimputed and the 31 imputed in total are right.

The count of disbanded and discredited cases. Step 1 names two democratization cases, Bolivia and the Philippines, then says in the next sentence that “the observed outcome $Y_i(0)$ in each of these four cases was 0”. Appendix Table B.2 lists two cases at that step. Appendix Table B.1 lists four end-of-conflict cases at the same step, which is where the count appears to have come from.

4.1 Two appendix cells the deposit does not support

Appendix Tables B.1 and B.2 print the full case datasets, 763 cells between them, and the deposited case file reproduces 761.

Burundi’s imputation probability. Table B.2 prints the imputed treated outcome for Burundi (1996-2005) as 1 with a probability of 0.9. Appendix C, which gives the reasoning case by case, states 0.8 for the same case, and the deposited data carry 0.8. The simulation uses 0.8, so the figures are unaffected; the table cell is the outlier.

Azerbaijan’s observed outcome. Table B.2 prints an observed outcome of 0 for Azerbaijan (1991-1992), consistent with the untreated potential outcome of 0 printed beside it. The deposited `outcome` column records 1 for that case while its `y0_obs` column records 0, so the deposit contradicts the switching equation for this one case. Nothing in the analysis reads the `outcome` column, so nothing published depends on it, and the table matches what the analysis used.

4.2 One probability the deposit carries as its complement

The South Africa case is imputed in the main text as $Y_i(0) = 1$ “with a probability of .8”, and appendix C repeats the 0.8. The deposited data carry 0.2, and appendix Table B.2 prints 0.2 alongside an imputed value of 1, which is the only case in either table where a point imputation of 1 sits beside a probability below one half. The simulation behind Figures 2 and 3 draws from 0.2.

The maintained rewrite does not change it, because choosing between the two is an analytical decision rather than a translation. Its consequences are small and worth stating: at 0.8 the final democratization bounds move from $[-2, 49]$ to $[-3, 48]$, the width stays at 51 points against 51, and every conclusion in the article is unaffected.

No other ground truth row is marked `match_rewrite = 0`.

5 Number-by-number comparison

Table 4: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it.

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
table_3	column d	1,1,1,1,1,1,0,0,0	1,1,1,1,1,1,0,0,0	1,1,1,1,1,1,0,0,0	1	1
table_3	column Y	1,1,1,0,0,0,1,0,0	1,1,1,0,0,0,1,0,0	1,1,1,0,0,0,1,0,0	1	1
table_3	column Y0_initial	?,?,?,?,?,?,1,0,0	?,?,?,?,?,?,1,0,0	?,?,?,?,?,?,1,0,0	1	1
table_3	column Y1_initial	1,1,1,0,0,0,?,?,?	1,1,1,0,0,0,?,?,?	1,1,1,0,0,0,?,?,?	1	1
table_3	column Y0_easy	1,1,?,?,0,?,?,1,0,0	1,1,?,?,0,?,?,1,0,0	1,1,?,?,0,?,?,1,0,0	1	1
table_3	column Y1_easy	1,1,1,0,0,0,1,0,?	1,1,1,0,0,0,1,0,?	1,1,1,0,0,0,1,0,?	1	1
table_3	column Y0_nulls	1,1,?,?,0,0,0,1,0,0	1,1,?,?,0,0,0,1,0,0	1,1,?,?,0,0,0,1,0,0	1	1
table_3	column Y1_nulls	1,1,1,0,0,0,1,0,?	1,1,1,0,0,0,1,0,?	1,1,1,0,0,0,1,0,?	1	1
table_3	column Y0_hard	1,1,0,0,0,0,1,0,0	1,1,0,0,0,0,1,0,0	1,1,0,0,0,0,1,0,0	1	1
table_3	column Y1_hard	1,1,1,0,0,0,1,0,?	1,1,1,0,0,0,1,0,?	1,1,1,0,0,0,1,0,?	1	1
table_3	EV bounds low, Initial Values	-40	-40	-40	1	1
table_3	EV bounds high, Initial Values	60	60	60	1	1
table_3	EV bounds low, Easy Cases	-20	-20	-20	1	1
table_3	EV bounds high, Easy Cases	30	30	30	1	1
table_3	EV bounds low, Nulls for 6&7	0	0	0	1	1
table_3	EV bounds high, Nulls for 6&7	30	30	30	1	1
table_3	EV bounds low, Hard Cases	20	20	20	1	1
table_3	EV bounds high, Hard Cases	30	30	30	1	1
figure_1	low bound, Before Any Data	-100	-100	-100	1	1
figure_1	high bound, Before Any Data	100	100	100	1	1
figure_1	low bound, Initial Values	-40	-40	-40	1	1
figure_1	high bound, Initial Values	60	60	60	1	1
figure_1	low bound, Easy Cases	-20	-20	-20	1	1
figure_1	high bound, Easy Cases	30	30	30	1	1
figure_1	low bound, Nulls for 6&7	0	0	0	1	1
figure_1	high bound, Nulls for 6&7	30	30	30	1	1
figure_1	low bound, Hard Cases	20	20	20	1	1
figure_1	high bound, Hard Cases	30	30	30	1	1
table_4	EV bounds low, unit 6 = 0, unit 7 = 0	0	0	0	1	1
table_4	EV bounds high, unit 6 = 0, unit 7 = 0	30	30	30	1	1
table_4	probability, unit 6 = 0, unit 7 = 0	0.56	0.56	0.56	1	1
table_4	EV bounds low, unit 6 = 1, unit 7 = 0	-10	-10	-10	1	1
table_4	EV bounds high, unit 6 = 1, unit 7 = 0	20	20	20	1	1
table_4	probability, unit 6 = 1, unit 7 = 0	0.14	0.14	0.14	1	1
table_4	EV bounds low, unit 6 = 0, unit 7 = 1	-10	-10	-10	1	1
table_4	EV bounds high, unit 6 = 0, unit 7 = 1	20	20	20	1	1
table_4	probability, unit 6 = 0, unit 7 = 1	0.24	0.24	0.24	1	1
table_4	EV bounds low, unit 6 = 1, unit 7 = 1	-20	-20	-20	1	1
table_4	EV bounds high, unit 6 = 1, unit 7 = 1	10	10	10	1	1
table_4	probability, unit 6 = 1, unit 7 = 1	0.06	0.06	0.06	1	1
table_4	probability-weighted point estimate, lower bound	-5		-5		1
table_4	probability-weighted point estimate, upper bound	25		25		1
table_4	2.5th quantile of the lower bound	-20		-20		1
table_4	97.5th quantile of the lower bound	0		0		1
table_4	2.5th quantile of the upper bound	10		10		1

Table 4: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it.
(continued)

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
table_4	97.5th quantile of the upper bound	30		30		1
table_4	probability that unit 6 has $Y_0 = 1$	0.3	0.2	0.2	0	0
table_4	probability that unit 7 has $Y_0 = 1$	0.4	0.3	0.3	0	0
table_5	democratization, Negative Effect	2% (1)	2% (1)	2% (1)	1	1
table_5	end of conflict, Negative Effect		0% (0)	0% (0)		
table_5	democratization, No Effect	40% (25)	40% (25)	40% (25)	1	1
table_5	end of conflict, No Effect		43% (23)	43% (23)		
table_5	democratization, Positive Effect	8% (5)	8% (5)	8% (5)	1	1
table_5	end of conflict, Positive Effect		13% (7)	13% (7)		
table_5	democratization, Unimputed	51% (32)	51% (32)	51% (32)	1	1
table_5	end of conflict, Unimputed		44% (24)	44% (24)		
figure_2	low bound, Before Any Data	-100	-100	-100	1	1
figure_2	high bound, Before Any Data	100	100	100	1	1
figure_2	low bound, Initial Values	-25	-25	-25	1	1
figure_2	high bound, Initial Values	75	75	75	1	1
figure_2	low bound, Disbanded and Discredited Cases	-25	-25	-25	1	1
figure_2	high bound, Disbanded and Discredited Cases	72	72	72	1	1
figure_2	low bound, Treated Cases	-16	-16	-16	1	1
figure_2	high bound, Treated Cases	68	68	68	1	1
figure_2	low bound, Non-transitional Cases	-12	-12	-12	1	1
figure_2	high bound, Non-transitional Cases	62	62	62	1	1
figure_2	low bound, Untreated Cases	-2	-2	-2	1	1
figure_2	high bound, Untreated Cases	49	49	49	1	1
figure_2	low bound, Unimputable Cases	-2	-2	-2	1	1
figure_2	high bound, Unimputable Cases	49	49	49	1	1
figure_a1	low bound, Before Any Data	-100	-100	-100	1	1
figure_a1	high bound, Before Any Data	100	100	100	1	1
figure_a1	low bound, Initial Values	-33	-33	-33	1	1
figure_a1	high bound, Initial Values	67	67	67	1	1
figure_a1	low bound, Disbanded and Discredited Cases	-32	-32	-32	1	1
figure_a1	high bound, Disbanded and Discredited Cases	61	61	61	1	1
figure_a1	low bound, Treated Cases	-26	-26	-26	1	1
figure_a1	high bound, Treated Cases	56	56	56	1	1
figure_a1	low bound, Non-transitional Cases	-24	-24	-24	1	1
figure_a1	high bound, Non-transitional Cases	54	54	54	1	1
figure_a1	low bound, Untreated Cases	-3	-3	-3	1	1
figure_a1	high bound, Untreated Cases	42	42	42	1	1
figure_a1	low bound, Unimputable Cases	-3	-3	-3	1	1
figure_a1	high bound, Unimputable Cases	42	42	42	1	1
figure_3	low bound, ATU, Before Any Data	-100	-100	-100	1	1
figure_3	high bound, ATU, Before Any Data	100	100	100	1	1
figure_3	low bound, ATU, Initial Values	-16	-16	-16	1	1
figure_3	high bound, ATU, Initial Values	84	84	84	1	1
figure_3	low bound, ATU, Disbanded and Discredited Cases	-16	-16	-16	1	1
figure_3	high bound, ATU, Disbanded and Discredited Cases	80	80	80	1	1
figure_3	low bound, ATU, Treated Cases	-16	-16	-16	1	1
figure_3	high bound, ATU, Treated Cases	80	80	80	1	1
figure_3	low bound, ATU, Non-transitional Cases	-12	-12	-12	1	1
figure_3	high bound, ATU, Non-transitional Cases	74	74	74	1	1
figure_3	low bound, ATU, Untreated Cases	0	0	0	1	1
figure_3	high bound, ATU, Untreated Cases	58	58	58	1	1
figure_3	low bound, ATU, Unimputable Cases	0	0	0	1	1
figure_3	high bound, ATU, Unimputable Cases	58	58	58	1	1
figure_3	low bound, ATT, Before Any Data	-100	-100	-100	1	1
figure_3	high bound, ATT, Before Any Data	100	100	100	1	1
figure_3	low bound, ATT, Initial Values	-88	-88	-88	1	1
figure_3	high bound, ATT, Initial Values	12	12	12	1	1
figure_3	low bound, ATT, Disbanded and Discredited Cases	-88	-88	-88	1	1
figure_3	high bound, ATT, Disbanded and Discredited Cases	12	12	12	1	1

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(continued)

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
figure_3	low bound, ATT, Treated Cases	-15	-15	-15	1	1
figure_3	high bound, ATT, Treated Cases	-15	-15	-15	1	1
figure_3	low bound, ATT, Non-transitional Cases	-15	-15	-15	1	1
figure_3	high bound, ATT, Non-transitional Cases	-15	-15	-15	1	1
figure_3	low bound, ATT, Untreated Cases	-15	-15	-15	1	1
figure_3	high bound, ATT, Untreated Cases	-15	-15	-15	1	1
figure_3	low bound, ATT, Unimputable Cases	-15	-15	-15	1	1
figure_3	high bound, ATT, Unimputable Cases	-15	-15	-15	1	1
figure_a2	low bound, ATU, Before Any Data	-100	-100	-100	1	1
figure_a2	high bound, ATU, Before Any Data	100	100	100	1	1
figure_a2	low bound, ATU, Initial Values	-29	-29	-29	1	1
figure_a2	high bound, ATU, Initial Values	71	71	71	1	1
figure_a2	low bound, ATU, Disbanded and Discredited Cases	-27	-27	-27	1	1
figure_a2	high bound, ATU, Disbanded and Discredited Cases	64	64	64	1	1
figure_a2	low bound, ATU, Treated Cases	-27	-27	-27	1	1
figure_a2	high bound, ATU, Treated Cases	64	64	64	1	1
figure_a2	low bound, ATU, Non-transitional Cases	-26	-26	-26	1	1
figure_a2	high bound, ATU, Non-transitional Cases	62	62	62	1	1
figure_a2	low bound, ATU, Untreated Cases	-2	-2	-2	1	1
figure_a2	high bound, ATU, Untreated Cases	48	48	48	1	1
figure_a2	low bound, ATU, Unimputable Cases	-2	-2	-2	1	1
figure_a2	high bound, ATU, Unimputable Cases	48	48	48	1	1
figure_a2	low bound, ATT, Before Any Data	-100	-100	-100	1	1
figure_a2	high bound, ATT, Before Any Data	100	100	100	1	1
figure_a2	low bound, ATT, Initial Values	-67	-67	-67	1	1
figure_a2	high bound, ATT, Initial Values	33	33	33	1	1
figure_a2	low bound, ATT, Disbanded and Discredited Cases	-67	-67	-67	1	1
figure_a2	high bound, ATT, Disbanded and Discredited Cases	33	33	33	1	1
figure_a2	low bound, ATT, Treated Cases	-10	-10	-10	1	1
figure_a2	high bound, ATT, Treated Cases	-10	-10	-10	1	1
figure_a2	low bound, ATT, Non-transitional Cases	-10	-10	-10	1	1
figure_a2	high bound, ATT, Non-transitional Cases	-10	-10	-10	1	1
figure_a2	low bound, ATT, Untreated Cases	-10	-10	-10	1	1
figure_a2	high bound, ATT, Untreated Cases	-10	-10	-10	1	1
figure_a2	low bound, ATT, Unimputable Cases	-10	-10	-10	1	1
figure_a2	high bound, ATT, Unimputable Cases	-10	-10	-10	1	1
figure_4	low bound, us, The 20 Cases with Expert Responses	-10	-10	-10	1	1
figure_4	high bound, us, The 20 Cases with Expert Responses	40	40	40	1	1
figure_4	low bound, expert, The 20 Cases with Expert Responses	-15	-15	-15	1	1
figure_4	high bound, expert, The 20 Cases with Expert Responses	15	15	15	1	1
figure_4	low bound, us, All 63 Democratization Cases	-2	-2	-2	1	1
figure_4	high bound, us, All 63 Democratization Cases	49	49	49	1	1
figure_4	low bound, expert, All 63 Democratization Cases	-19	-19	-19	1	1
figure_4	high bound, expert, All 63 Democratization Cases	59	59	59	1	1
figure_4	low bound, combined, All 63 Democratization Cases	-3	-3	-3	1	1
figure_4	high bound, combined, All 63 Democratization Cases	41	41	41	1	1
abstract	bounds width prior to any analysis	100	100	100	1	1
abstract	bounds width after imputation	51	51	51	1	1
abstract	bounds width after the expert survey	44	44	44	1	1
text	democratization cases	63	63	63	1	1
text	treated democratization cases	8	8	8	1	1

Table 4: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it.
(continued)

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
text	bounds width before any data are collected	200	200	200	1	1
text	bounds width once half the potential outcomes are revealed	100	100	100	1	1
text	final ATE lower bound	-2	-2	-2	1	1
text	final ATE upper bound	49	49	49	1	1
text	final ATE bounds width	51	51	51	1	1
text	toy example: treated units with outcome 1	3	4	4	0	0
text	toy example: untreated units with outcome 1	1	1	1	1	1
text	toy example: bounds width before any data	200	200	200	1	1
text	toy example: final bounds width	10	10	10	1	1
text	ATU final bounds width	58	58	58	1	1
text	ATT summarised as a point effect	-15	-15	-15	1	1
text	cases imputed as a non-zero causal effect	6	6	6	1	1
text	expert responses received	20	20	20	1	1
text	democratization cases without an expert response	43	43	43	1	1
text	our bounds width, 20 responding cases	50	50	50	1	1
text	expert bounds width, 20 responding cases	30	30	30	1	1
text	our bounds width, all 63 cases	51	51	51	1	1
text	expert bounds width, all 63 cases	78	78	78	1	1
text	combined bounds width, all 63 cases	44	44	44	1	1
text	expert and author imputations agreed	7	7	7	1	1
text	expert and author imputations conflicted	3	3	3	1	1
text	we imputed, the experts did not	3	3	3	1	1
text	the experts imputed, we did not	7	7	7	1	1
text	cases the experts imputed	14	14	14	1	1
text	cases we imputed	10	10	10	1	1
text	experts who declined to impute	6	6	6	1	1
text	declining experts assigned to an untreated case	6	6	6	1	1
text	expert imputations differing from the observed outcome	0	0	0	1	1
appendix_a	end-of-conflict cases	54	54	54	1	1
appendix_a	treated end-of-conflict cases	6	6	6	1	1
appendix_a	bounds width once half the potential outcomes are revealed	100	100	100	1	1
appendix_a	bounds width after imputation	41	44	44	0	0
appendix_a	final ATE lower bound	-3	-3	-3	1	1
appendix_a	final ATE upper bound	42	42	42	1	1
appendix_a	final ATE bounds width from the rounded endpoints	45	45	45	1	1
appendix_a	ATT summarised as a point effect	-10	-10	-10	1	1
text	toy example: units	10		10		1
text	toy example: treated units	7		7		1
text	toy example: untreated units	3		3		1
text	toy example: easy imputations	5		5		1
text	toy example: untreated outcomes for units 1, 2 and 5	1, 1, 0		1, 1, 0		1
text	toy example: treated outcomes for units 8 and 9	1, 0		1, 0		1
text	probabilistic extension: possible sets of potential outcomes	4		4		1
text	unobserved potential outcomes	63		63		1
text	unobserved potential outcomes imputed with certainty	5		4		0
text	unobserved potential outcomes imputed probabilistically	26		27		0
text	unobserved potential outcomes left unimputed	32		32		1
text	disbanded and discredited democratization cases	4		2		0
text	observed outcome in the disbanded and discredited cases	0		0		1
text	South Africa: probability the imputed untreated outcome equals 1	0.8		0.2		0
text	disbanded cases: probability the imputed outcome equals 1	0.1		0.1		1

Table 4: Ground truth: the published value against the deposited script and against the maintained rewrite. A blank Paper column means the quantity is not stated in the article or its appendix; a blank Archive column means no deposited script prints it. *(continued)*

Location	Quantity	Paper	Archive	Rewrite	Match	Match (rw)
text	Nigeria: probability the imputed untreated outcome equals 1	0.1		0.1		1
text	experts disagreeing with our coding of treatment and outcome	5				
text	expert imputations conflicting with ours, restated in the discussion	3		3		1
text	end-of-conflict cases, stated in the main text footnote	54		54		1
text	unimputed cases as a share of the total			32 of 63, 50.8%		1
text	the bounds around the ATE include zero			[-2, 49] contains 0		1
text	the ATT bounds shrink to a point			[-15, -15], width 0		1
text	the ATU bounds are wider than the ATT bounds			58 against 0		1
text	the combined bounds are narrower than our original bounds			44 against 51		1
text	no expert assigned to a treated case declined to impute			0 of 6 declining experts		1
text	expert bounds are narrower than ours among the 20 responding cases			30 against 50		1
text	expert bounds are wider than ours among all 63 cases			78 against 51		1
table_b1	cells printed in the treatment indicator column	54		54		1
table_b1	cells printed in the observed outcome column	54		54		1
table_b1	cells printed in the revealed potential outcome column	54		54		1
table_b1	cells printed in the imputed untreated outcome column	54		54		1
table_b1	cells printed in the imputed treated outcome column	54		54		1
table_b1	cells printed in the imputation probability column	30		30		1
table_b1	cells printed in the unit-level effect column	54		54		1
table_b2	cells printed in the treatment indicator column	63		63		1
table_b2	cells printed in the observed outcome column	63		62		0
table_b2	cells printed in the revealed potential outcome column	63		63		1
table_b2	cells printed in the imputed untreated outcome column	63		63		1
table_b2	cells printed in the imputed treated outcome column	63		63		1
table_b2	cells printed in the imputation probability column	31		30		0
table_b2	cells printed in the unit-level effect column	63		63		1

Of the 232 recorded claims, 181 can be compared against both the article and a deposited script, and 177 of those match. Six further claims are stated in the article but computed by no deposited script: the probability-weighted point estimate of the bounds in the probabilistic extension, [-5, 25], and the 2.5th and 97.5th quantiles of each bound, [-20, 0] and [10, 30]. The rewrite computes all six from the enumerated scenarios in `table_4_probabilistic.R` and all six match.

5.1 Coverage

The list of numbers to check is built from the article rather than from the pipeline, because a list built from the pipeline is silent exactly where the pipeline is missing something.

`ground_truth/published_claims.csv` is every numeric token the article and its supporting information print, 268 of them, each classified by hand. 228 are quantities this pipeline can move, and each of those must be checked twice: once by a row in the ground truth and once by a block in `maintained/in_text_claims.R`, which reaches the same number by its own path through the same output files. The remaining 40 are scale endpoints, unit indices, period ranges and values copied out of other authors' papers, none of which any analysis can change.

Table 5: The published claims extraction, by class.

Class	Claims	Checked by
definitional	22	at the point of use
descriptive	9	ground truth and <code>in_text_claims.R</code>
pipeline	219	ground truth and <code>in_text_claims.R</code>
structural	14	at the point of use
transcribed	4	at the point of use

`ground_truth/build_ground_truth.R` enforces this. It sources the claims file rather than reading it as text, because a block that errors or prints nothing satisfies a textual check completely, and it counts the printed claims against the extraction. It also compares the two instruments value by value: they filter, convert and round independently, so a disagreement between them means one of the two is wrong. The run halts on any of these.

5.2 Appendix Tables B.1 and B.2

Appendix B prints the full case datasets, 54 end-of-conflict cases and 63 democratization cases, with seven columns each. The deposited archive has no script for either table, so until `table_b1_b2_case_dataset.R` was written the 763 published cells had nothing on this side to be compared against. All but 2 reproduce; both exceptions are described in the errata section above.

6 Maintained rewrite

The rewrite lives in `maintained/`: a helpers file, one cleaning script, three table scripts, four figure scripts and three in-text scripts. It is a translation, not a reanalysis. `ev_bounds()`, `bounds_width()`, `sample_bounds()` and the probabilistic-extension functions are the paper's contribution and are carried over with their logic unchanged; what changes is the tidyverse around them.

6.1 Architecture

Table 6: The maintained rewrite.

Script	Output
helpers.R	none; packages, the QUIMPO functions and the toy example
clean_cases.R	cases_long.rds
table_3_toy_example.R	table_3_toy_example.csv, table_3_bounds.csv
table_4_probabilistic.R	table_4_probabilistic.csv, table_4_summary.csv
table_5_imputation_summary.R	table_5_imputation_summary.csv
table_b1_b2_case_dataset.R	table_b1_b2_case_dataset.csv
figure_1_toy_bounds.R	figure_1_toy_bounds.pdf/.png/.csv
figures_2_a1_ate_bounds.R	figure_2_dem_ate_bounds.pdf/.png, figure_a1_eoc_ate_bounds.pdf/.png, figures_2_a1_gg_df.csv
figures_3_a2_att_atu_bounds.R	figure_3_dem_att_atu_bounds.pdf/.png, figure_a2_eoc_att_atu_bounds.pdf/.png, figures_3_a2_gg_df.csv
figure_4_expert_validation.R	figure_4_expert_validation.pdf/.png/.csv
text_expert_agreement.R	text_expert_agreement.csv, text_expert_agreement_counts.csv
text_imputation_counts.R	text_imputation_counts.csv
text_summary_stats.R	text_summary_stats.csv
text_seed_sensitivity.R	text_seed_sensitivity.csv
in_text_claims.R	none; one printed line per published claim, checked by build_ground_truth.R

Three departures from a line-by-line translation are worth naming.

The toy example is defined once, in `quimpo_toy_example()` in `helpers.R`, rather than retyped in each of the three scripts that need it. In the archive, `table_3_figure_1.R` and `table_4.R` each set up the ten units and walk them through the imputation steps independently, which is the arrangement in which Table 3, Figure 1 and Table 4 can silently come to describe different examples.

Figures 2 and A1 come from one script and Figures 3 and A2 from another, because in each pair the two figures are two facets of a single simulation and splitting them would mean running it twice at the same seed. The archive’s `figures_2_3_A1_A2.R` puts all four in one file; the rewrite splits along the simulation boundary rather than the file boundary.

`text_summary_stats.R` reads the bounds back out of the figure output rather than recomputing them, so the widths quoted in the abstract and the body cannot drift from the figures they summarise. Every number it prints is traceable to a committed CSV, and no published value is typed into any script in `maintained/`.

6.2 Deprecated patterns replaced

Table 7: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
<code>'rm(list = ls())'</code>	(omitted)
commented-out <code>'setwd()'</code>	<code>'here::here()'</code>

<pre> magrittr pipe 'reshape2::melt()' / 'dcast()' 'do(sample_bounds(...))' 'map_df(enframe, .id =)' 'geom_errorbarh(height =)' 'geom_linerange(size =)' '<ggplot> %+% data' 'separate()' 'recode_factor()' 'xtable' + 'print.xtable' (commented out) 'ggsave()' (commented out) unseeded 'rbinom()' simulation bounds printed as fractions in 'table_4.R' </pre>	<pre> native pipe 'pivot_longer()' / 'pivot_wider()' 'reframe(sample_bounds(...))' 'map(enframe) &#124;> bind_rows(.id =)' 'geom_errorbar(orientation = "y", width =)' 'geom_linerange(linewidth =)' a named plot function taking the data as an argument 'separate_wider_delim()' 'case_when()' + 'factor()' 'write_csv()' to 'output/' 'ggsave()' to 'output/', PDF and PNG 'set.seed(12345)' multiplied by 100, as the paper prints them </pre>
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7 Figures

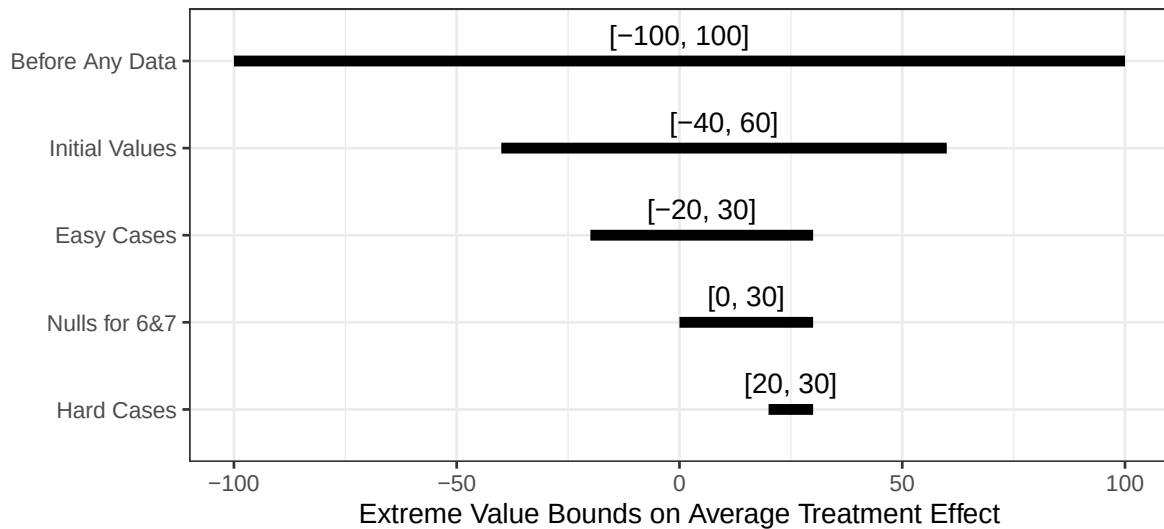


Figure 1: Figure 1: extreme value bounds at each step of the toy example.

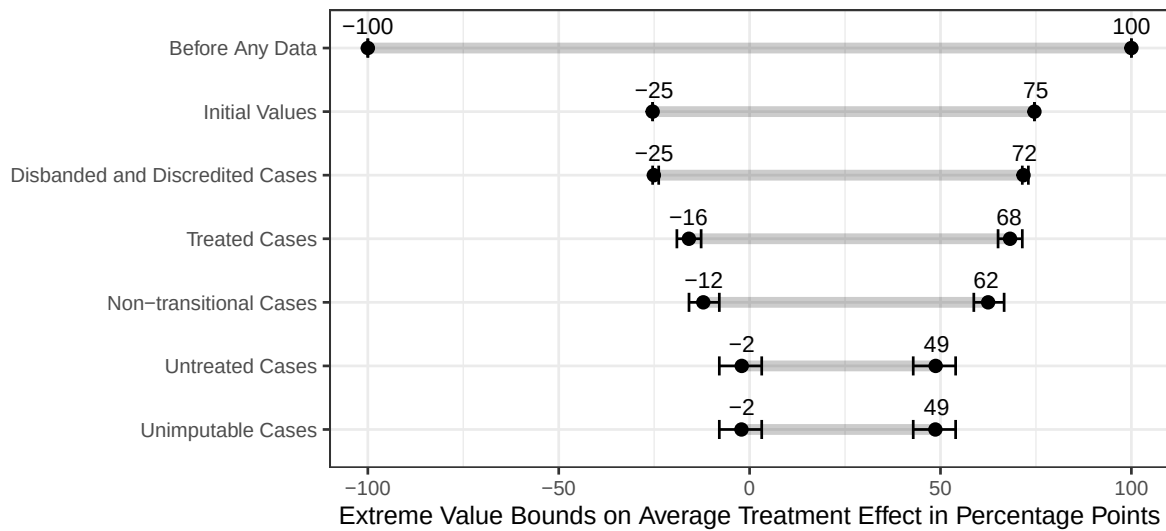


Figure 2: Figure 2: transitional truth commissions, extreme value bounds on the ATE.

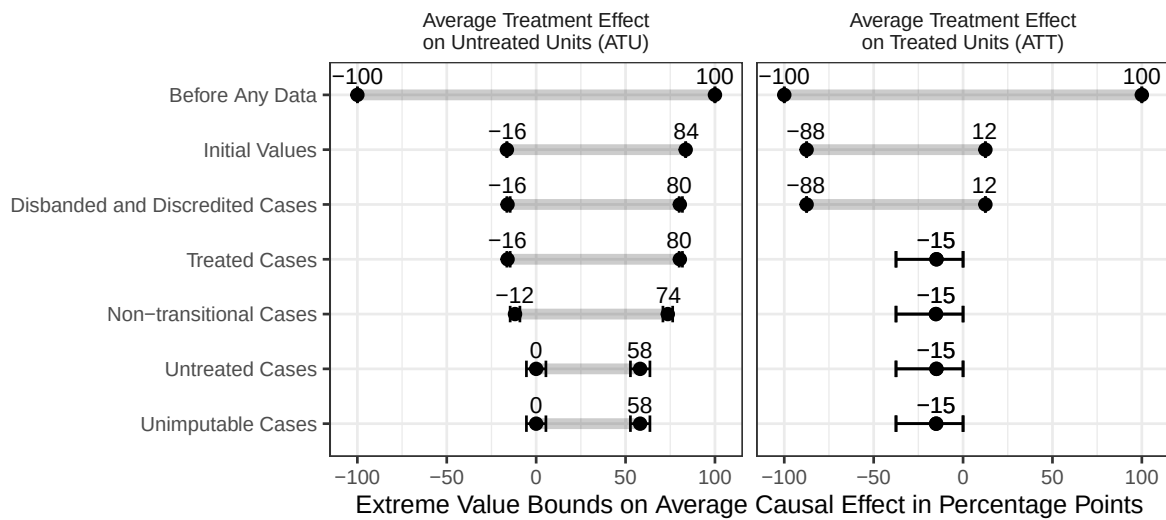


Figure 3: Figure 3: extreme value bounds by treatment status.

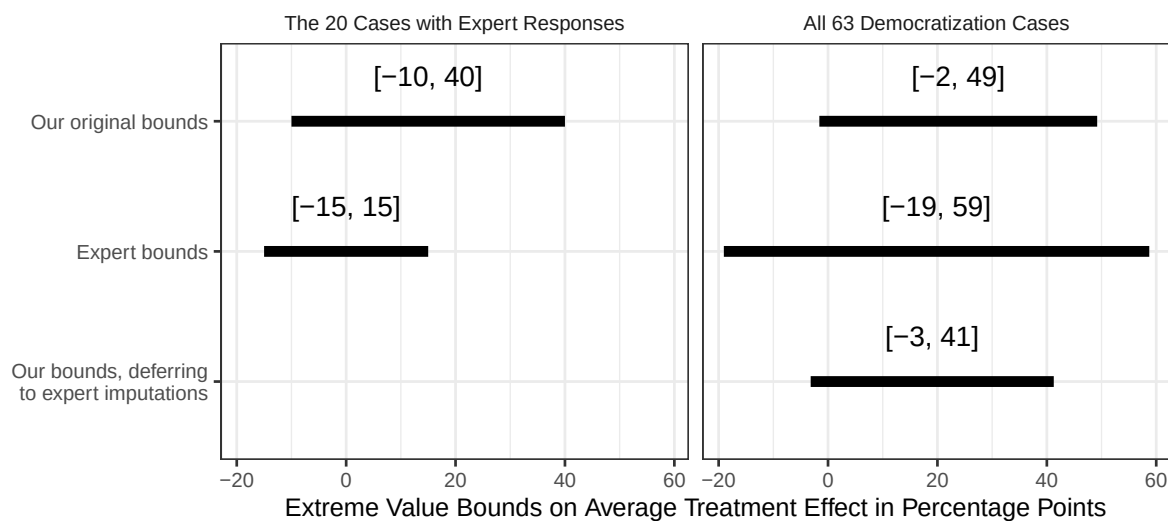


Figure 4: Figure 4: extreme value bounds implied by the expert imputations.

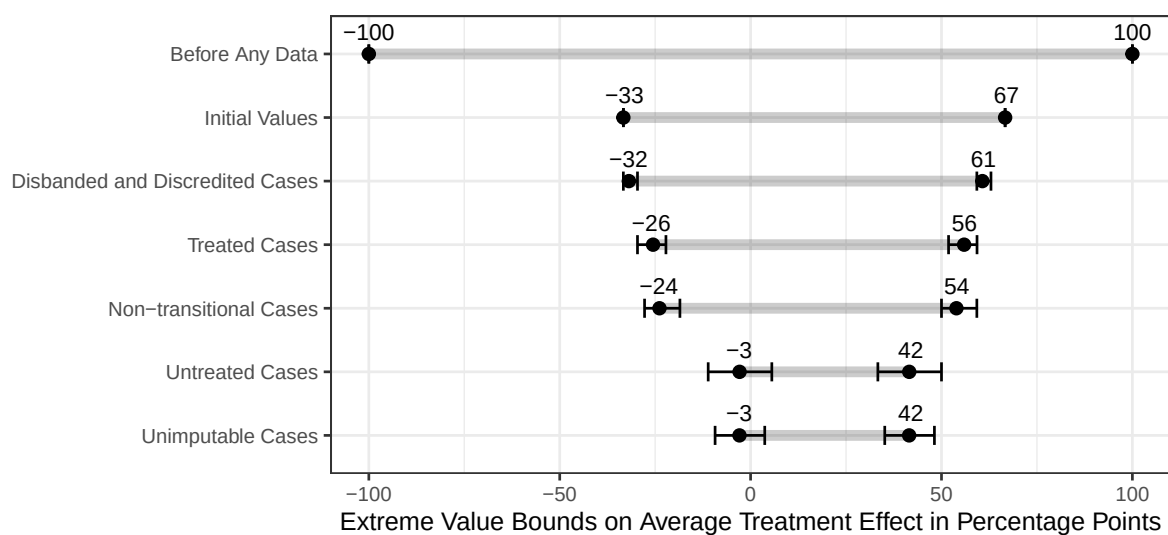


Figure 5: Figure A1: end-of-conflict cases, extreme value bounds on the ATE.

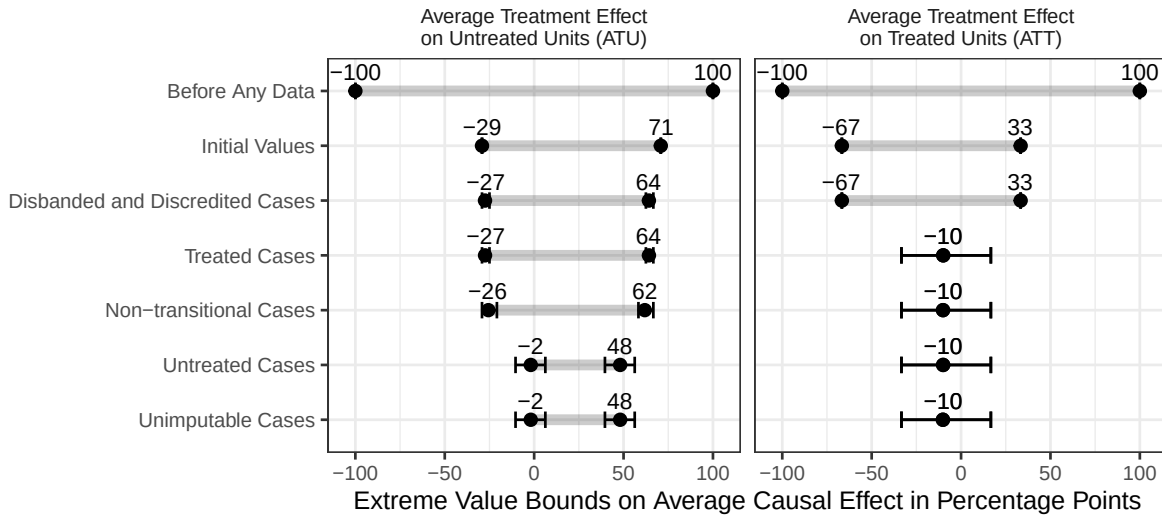


Figure 6: Figure A2: end-of-conflict cases, extreme value bounds by treatment status.

8 Expert survey verification

The Expert Survey section carries eleven quantities that no deposited script writes to disk, three of them recorded only in a code comment in `figure_4.R`. `text_expert_agreement.R` recomputes all of them from the deposited case data.

Table 8: The Expert Survey section, recomputed. Agreement means the two sides made the same imputation or both declined; the last row checks the claim that all 20 respondents either declined or imputed the observed outcome.

Quantity	Rewrite	Paper
agreement	7	7
disagreement	3	3
They declined, we imputed	3	3
They imputed, we declined	7	7
expert responses received	20	20
democratization cases without an expert response	43	43
cases the experts imputed	14	14
cases we imputed	10	10
experts who declined to impute	6	6
declining experts assigned to an untreated case	6	6
expert imputations differing from the observed outcome	0	0

The archive’s `figure_4.R` computes the agreement coding and prints it with `table()`, then records the answer in three comment lines: “uruguay, burundi, south africa are ‘disagreements’”,

“7 cases, they imputed when we did not”, “3 cases, we imputed and they did not”. All three are correct. They are also the only place the numbers exist, which is the pattern this program treats as a defect regardless of whether the comment is right: a comment is not an output, and nothing checks it.

9 In-text quantities

Table 9: Bounds widths and summary effects quoted in the abstract, the body and Appendix A, read back out of the figure output that produced them.

Quantity	Rewrite	Rounded
dem: bounds width before any data	200.00	200
dem: bounds width once the world reveals half the potential outcomes	100.00	100
dem: final ATE lower bound	-2.14	-2
dem: final ATE upper bound	48.66	49
dem: final ATE bounds width	50.79	51
dem: ATU final bounds width	58.18	58
dem: ATT final lower bound	-15.05	-15
dem: ATT final upper bound	-15.05	-15
expert: our bounds width, 20 responding cases	50.00	50
expert: expert bounds width, 20 responding cases	30.00	30
expert: our bounds width, all 63 cases	50.79	51
expert: expert bounds width, all 63 cases	77.78	78
expert: combined bounds width, all 63 cases	44.44	44
dem: cases imputed as a non-zero causal effect	6.00	6
eoc: bounds width once the world reveals half the potential outcomes	100.00	100
eoc: final ATE lower bound	-2.91	-3
eoc: final ATE upper bound	41.53	42
eoc: final ATE bounds width	44.44	44
eoc: ATT final lower bound	-10.14	-10
eoc: ATT final upper bound	-10.14	-10

The abstract and the body describe the same starting point in two ways, and both are right. The body says the bounds are 200 points wide “before any data are collected” and shrink to 100 “after the world reveals half the potential outcomes”. The abstract skips the first of those and says the bounds are 100 points wide “prior to any analysis”, meaning after data collection and before imputation. Both numbers are in the pipeline and both reproduce.

10 R environment

Table 10: Package versions used for this report.

Package	Version
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tidyverse	2.0.0
here	1.0.2
janitor	2.2.1
knitr	1.51
kableExtra	1.4.0
ggplot2	4.0.3
dplyr	1.2.1
tidyr	1.3.2

R version: R version 4.6.0 (2026-04-24). The archive was written against R 4.1.2.